Data preprocessing and feature engineering

* Theory

1) Write a short note. what is data analysis?

→ Data analysis is a process of collecting, cleaning Transforming, exploring and interpreting data to find useful information and support decision-Making.

* The Main steps are:
1. Collect data
2. Understand data
3. Clean data
4. Transform data
5. Analyze data
6. Visualize data
7. Find insights
8. Make decisions.

→ Raw customer data
↓
clean the data
↓
Analyze purchases
↓
Find customer patterns
↓
Predict customer churn

2. Describe how to plan a data science project, Listing all steps.

→ 1. problem definition
2. Data collection
3. Data understanding
4. Data cleaning
5. Data preprocessing
6. Exploratory data analysis (EDA)
7. Feature engineering
8. Model Building
9. Model evaluation
10. Deployment
11. Monitoring

3. Frame a Machine Learning problem statement Predict whether a customer will churn based on purchase behaviour

→ not a clear problem; we want to understand customer

→ framed problem: Predict whether a customer will churn based on demographic information and purchasing behaviour.

→ Input columns (Explanatory variables)
→ Age
→ Gender
→ income
→ Purchases
→ Total spent

→ Output

4) Explain
→ what
→ pro
simple

→ A Te
vscol

→ Output (target column) → Churn

4) Explain :
→ what are tensors
→ provide an in-depth explanation of tensors with simple example using numpy

→ A Tensor is a general Mathematical structure used to store data in Multiple dimensions..